Statistics for the Overall Findings Storyboard (overall_v1)

Colter Bay, Grand Teton National Park, 2022

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All percentages are back-transformed from the log scale: % change in detections per km away from the parking lot, or % difference under white vs. red light. 95% CIs are Wald intervals; p-values are Wald z-tests. "Holm" and "BH" mark p-values adjusted across the five intensities (Holm) or across the taxa tested (Benjamini–Hochberg). Light color and intensity were set experimentally; distance is observational.

Model formulas

All models are fitted with `glmmTMB` (negative binomial, `nbinom2`). `site` is a random intercept; light `color` (R/W) and `intensity` (10, 30, 50, 70, 100%) are factors; `dist_km` = distance from the parking lot; `jd_c` = day of year, centered.

Figures 1B–C, 2 and 5B–C — one model per taxon (Figures 1–2: 11 stations; Figure 5: 13 stations, GRTE15 excluded). Figure 1 uses the color × distance slopes; Figure 2 uses the intensity × distance slopes (A–B) and the color × intensity contrasts (C–D).

Bats (total detections per site-night; no zero-inflation):

```
total_det ~ color:intensity + intensity:dist_km + color:dist_km + color + intensity
  + dist_km + mean_phase + pct_nonforest + brightness_dark + jd_c + I(jd_c^2)
  + (1 | site)
```

Moths (specimens per trap night; no zero-inflation):

```
detections ~ color:intensity + intensity:dist_km + color:dist_km + color + intensity
  + dist_km + mean_phase + pct_nonforest + jd_c + I(jd_c^2) + brightness_dark
  + (1 | site)
```

Figure 3 — bat hierarchy (zero-inflated, `ziformula = ~1`; zero-filled site-nights). All bats uses total detections; guilds and species use their own detections with `taxon` = guild or species. Every tier converged with the full covariate set.

```
All bats:          total_det ~ dist_km + color + intensity + mean_phase + pct_nonforest
                     + brightness_dark + jd_c + I(jd_c^2) + (1 | site)
Guilds / species:  detections ~ taxon * dist_km + color + intensity + mean_phase
                     + pct_nonforest + brightness_dark + jd_c + I(jd_c^2) + (1 | site)
```

Figure 4 — moth hierarchy (zero-inflated, `ziformula = ~1`; one model per taxonomic level over the taxa with at least 30 site-nights, zero-filled trap nights). If a level's model did not converge, covariates were dropped in this order: non-forest cover, sky brightness, season^2, moon phase.

```
detections ~ taxon * dist_km + color + intensity + mean_phase + pct_nonforest
  + brightness_dark + jd_c + I(jd_c^2) + (1 | site)
```

In both hierarchies a node's slope and p-value come from the taxon-specific `dist_km` slope (`emtrends`); curves are averaged over color and intensity.

Figure 1A and 5A: Station sky brightness

Table 1: Sky brightness from the fisheye photos (dark sky units). Lights off = darkest lights-off photo (as used in the models); white 100% = mean of three photos (GRTE11 p1 excluded: transient light). Distance from the reference point at GRTE17.

Station	Distance (km)	Lights off (DSU)	White 100% (DSU)	Mapped (DSU)	In Figure 1
GRTE17 (lot)	0.00	2.9	28.6	28.6	yes
GRTE16 (lot)	0.37	15.3	83.2	83.2	yes
GRTE05	0.49	8.4	11.9	11.9	yes
GRTE01 (lot)	0.51	5.9	20.1	20.1	yes
GRTE06	0.77	3.1	3.6	3.6	yes
GRTE02	0.98	4.1	3.5	3.5	yes
GRTE08	1.56	4.3	2.2	2.2	yes
GRTE03	1.60	3.7	3.9	3.9	yes
GRTE09	1.92	4.0	—	4.0*	yes
GRTE11	2.02	4.3	4.9	4.9	yes
GRTE10	2.10	2.7	—	2.7*	yes
GRTE15	7.20	3.3	3.6	3.6	no (far)
GRTE14	7.40	24.0	9.6	9.6	no (far)
GRTE13	7.46	2.8	3.0	3.0	no (far)

* No white-light photos; the lights-off value is mapped instead. GRTE15 appears in neither map. The surface in the maps is inverse-distance-weighted interpolation of log DSU (power 2), clipped to 300 m around the stations.

Figure 1B–C: Distance slopes by light color (11 stations, 0–2.1 km)

Table 2: Models (negative binomial GLMMs; site random intercept).

Model	Obs.	Stations	Family	Converged
Bats: total detections per site-night	502	11	nbinom2	yes
Moths: specimens per trap night	176	11	nbinom2	yes

Both models: color × intensity, intensity × distance, color × distance, moon phase, non-forest cover, sky brightness (lights off), season (day of year, linear + quadratic) and a site random intercept. Full formulas: see *Model formulas* above.

Table 3: Figure 1B — bats: % change in detections per km under each light color (not adjusted).

Light	% per km	95% CI	z	p
Red	-53%	-77 to -7%	-2.16	0.030
White	-53%	-76 to -5%	-2.12	0.034

Table 4: Figure 1C — moths: % change in specimens per km under each light color (not adjusted).

Light	% per km	95% CI	z	p
Red	+3%	-21 to +33%	0.20	0.843
White	+63%	+27 to +109%	3.87	< 0.001

Figure 2: Effects at each light intensity (same models as Figure 1)

Table 5: Figure 2A — bats: % change in detections per km away from the lot, by intensity.

Intensity	Estimate	95% CI	z	p	p (Holm)
10%	-50%	-75 to -0%	-1.97	0.049	0.105
30%	-42%	-72 to +18%	-1.51	0.132	0.132
50%	-53%	-77 to -5%	-2.11	0.035	0.105
70%	-56%	-78 to -11%	-2.30	0.021	0.086
100%	-61%	-80 to -20%	-2.59	0.010	0.048

Table 6: Figure 2B — moths: % change in specimens per km away from the lot, by intensity.

Intensity	Estimate	95% CI	z	p	p (Holm)
10%	-4%	-35 to +41%	-0.21	0.831	1.000
30%	+25%	-16 to +85%	1.11	0.266	0.797
50%	+76%	+21 to +155%	2.98	0.003	0.011
70%	+74%	+27 to +138%	3.45	< 0.001	0.003
100%	-1%	-30 to +39%	-0.08	0.936	1.000

Table 7: Figure 2C — bats: % difference under white vs. red light, by intensity (at the mean distance).

Intensity	Estimate	95% CI	z	p	p (Holm)
10%	+6%	-18 to +36%	0.44	0.657	0.657
30%	-51%	-66 to -31%	-4.03	< 0.001	< 0.001
50%	-40%	-55 to -20%	-3.54	< 0.001	0.002
70%	-34%	-49 to -16%	-3.28	0.001	0.003
100%	+51%	+16 to +97%	3.03	0.002	0.005

Table 8: Figure 2D — moths: % difference under white vs. red light, by intensity (at the mean distance).

Intensity	Estimate	95% CI	z	p	p (Holm)
10%	-3%	-47 to +76%	-0.11	0.909	1.000
30%	-41%	-67 to +6%	-1.76	0.078	0.310
50%	-27%	-55 to +20%	-1.23	0.219	0.658
70%	+5%	-30 to +58%	0.23	0.817	1.000
100%	+190%	+87 to +351%	4.74	< 0.001	< 0.001

Red-light nights at 100% intensity: moth traps 4 nights (12–13 July, 5–6 August 2022), bat detectors 6 nights (11–13 July, 4–6 August); they include the July full moon and the two wettest nights of the study.

Figure 3: Bat distance hierarchy (11 stations)

Table 9: Figure 3 — distance slope at each node. All bats: single test (unadjusted); guilds and species: Benjamini–Hochberg across the 12 nodes.

Tier	Taxon	% per km	95% CI	z	p (raw)	p (adj.)
All bats	All bats	-53%	-76 to -10%	-2.29	0.022	0.022
Guild	Low frequency	-32%	-69 to +48%	-0.97	0.334	0.401
Guild	Myev	-45%	-75 to +21%	-1.49	0.136	0.204
Guild	High frequency	-60%	-81 to -12%	-2.30	0.022	0.065
Species	Epfu	-79%	-91 to -48%	-3.40	< 0.001	0.005
Species	Laci	+15%	-53 to +180%	0.31	0.758	0.758
Species	Lano	+18%	-51 to +187%	0.37	0.712	0.758
Species	Myca	-80%	-92 to -47%	-3.23	0.001	0.005
Species	Myci	-56%	-82 to +9%	-1.77	0.076	0.152
Species	Myev	-45%	-78 to +34%	-1.33	0.185	0.247
Species	Mylu	-60%	-83 to -1%	-1.99	0.046	0.111
Species	Myvo	-78%	-91 to -46%	-3.30	< 0.001	0.005
Species	Myyu	-53%	-81 to +19%	-1.59	0.112	0.192

Zero-inflated negative binomial GLMMs of detections per site-night (all bats = total; guild and species = their detections, zero-filled) with taxon × distance, light color, intensity, moon phase, non-forest cover, sky brightness, season and a site random intercept. Covariates were dropped in order if a model did not converge; covariates used: all bats — mean_phase + pct_nonforest + brightness_dark + jd_c + I(jd_c^2); guilds — mean_phase + pct_nonforest + brightness_dark + jd_c + I(jd_c^2); species — mean_phase + pct_nonforest + brightness_dark + jd_c + I(jd_c^2).

Figure 4: Moth distance hierarchy (11 stations)

Table 10: Figure 4 — the 17 taxa shown: Benjamini–Hochberg-adjusted $p < 0.05$ across all 51 taxa tested.

Level	Taxon	% per km	p (raw)	p (BH)
Family	Erebidae	+100%	< 0.001	< 0.001
Family	Lasiocampidae	+105%	< 0.001	< 0.001
Family	Sphingidae	+102%	< 0.001	0.001
Family	Notodontidae	+66%	0.014	0.043
Subfamily	Smerinthinae	+168%	< 0.001	< 0.001
Subfamily	Arctiinae	+113%	< 0.001	< 0.001
Subfamily	Sterrhinae	+170%	< 0.001	< 0.001
Subfamily	Lasiocampinae	+108%	< 0.001	0.001
Genus	Smerinthus	+153%	< 0.001	0.002
Genus	Apantesis	+148%	< 0.001	0.002
Genus	Scopula	+149%	0.002	0.009
Genus	Malacosoma	+122%	0.003	0.010
Genus	Feltia	+108%	0.014	0.043
Species	Apamea inficita	+188%	< 0.001	0.004
Species	Smerinthus ophthalmica	+154%	0.001	0.006
Species	Apantesis williamsii	+141%	0.003	0.011
Species	Malacosoma californica	+133%	0.009	0.029

Taxon × distance zero-inflated negative binomial GLMM at each taxonomic level (taxa with at least 30 site-nights), with light color, intensity, moon phase, non-forest cover, sky brightness, season and a site random intercept.

Table 11: All other taxa tested (not significant after adjustment).

Level	Taxon	% per km	p (raw)	p (BH)
Family	Noctuidae	+34%	0.071	0.157
Family	Geometridae	+4%	0.819	0.836
Subfamily	Noctuinae	+37%	0.067	0.157
Subfamily	Sphinginae	-42%	0.114	0.199
Subfamily	Erebinae	+35%	0.175	0.279
Subfamily	Pantheinae	-30%	0.251	0.366
Subfamily	Ennominae	-15%	0.325	0.417
Genus	Euxoa	+81%	0.021	0.060
Genus	Xestia	+111%	0.028	0.076
Genus	Apamea	+58%	0.058	0.148
Genus	Arctia	-48%	0.069	0.157
Genus	Leucania	+59%	0.080	0.169
Genus	Spilosoma	+72%	0.086	0.176
Genus	Trichordestra	-36%	0.098	0.193
Genus	Pero	-37%	0.109	0.199
Genus	Sphinx	-47%	0.112	0.199
Genus	Drasteria	+42%	0.217	0.336
Genus	Panthea	-32%	0.263	0.373
Genus	Agrotis	+29%	0.327	0.417
Genus	Polia	+34%	0.319	0.417
Genus	Macaria	-19%	0.375	0.466
Genus	Egira	+16%	0.518	0.587
Genus	Orthosia	+6%	0.836	0.836
Species	Spilosoma vagans	+71%	0.117	0.199
Species	Sphinx vashti	-47%	0.139	0.229
Species	Trichordestra liquida	-30%	0.243	0.364
Species	Panthea virginarius	-33%	0.302	0.416
Species	Agrotis vancouverensis	+28%	0.457	0.534
Species	Apamea devastator	+30%	0.442	0.534
Species	Egira curialis	+22%	0.461	0.534
Species	Pero behrensaria	-19%	0.562	0.623
Species	Macaria adonis	+11%	0.699	0.758
Species	Leucania commoides	+10%	0.778	0.827
Species	Apamea unanimis	+8%	0.806	0.836

Figure 5B–C: Distance slopes with the far stations (13 stations, 0–7.5 km; GRTE15 excluded)

Table 12: Models (negative binomial GLMMs; site random intercept).

Model	Obs.	Stations	Family	Converged
Bats: total detections per site-night	593	13	nbinom2	yes
Moths: specimens per trap night	210	13	nbinom2	yes

Table 13: Figure 5B — bats: % change in detections per km under each light color (not adjusted).

Light	% per km	95% CI	z	p
Red	-2%	-20 to +20%	-0.18	0.859
White	-2%	-20 to +19%	-0.23	0.816

Table 14: Figure 5C — moths: % change in specimens per km under each light color (not adjusted).

Light	% per km	95% CI	z	p
Red	+1%	-5 to +9%	0.38	0.706
White	+19%	+12 to +27%	5.32	< 0.001

Same model structure as Figure 1B–C, fitted on the 13 stations.